MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

INVERTER, DIGITAL, KEY CODE LDC

This amendment forms a part of MIL-M-28787/20A(EC), dated 22 March 1979, and is approved for use within the Space and Naval Warfare Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2, first line: Delete "100°C" and substitute "85°C".

1.2, Output voltage, Low level: Delete "0.V" and substitute "0.4 V".

PAGE 3

3.3, delete and substitute:

"3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.18 failures per million operating hours for class I or 0.45 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.4)."

3.4, delete and substitute:

"3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.4)."

PAGE 4

4.3, delete and substitute:

"4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

PAGES 6, 7, AND 8

TABLE II: Delete and substitute new table II as printed on pages 3 and 4 of this amendment.

PAGE 9

* 4.4.2, delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. Test conditions A, B, C, H and U are stress tests and will be performed only at 25°C ±5°C. If test conditions A, B, C, H, or U are part of a sequence of tests, the order of testing shall be as follows: U, A, B, C, H where applicable, followed by the remaining tests in the sequence."

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:29Z
Check the source to verify that this is the current version before use.

* 4.5.1, line five: Delete "C through N," and substitute "D through G, J through N,".

* 4.5.2, line four: Delete "column" and substitute "column (except test condition H),.

* 4.5.3, line three: Delete "C through R" and substitute "D through G, J through R".

* 4.5.3, line five: Delete "C through W" and substitute "D through G, J through T, V, and W".

6.3, delete and substitute:

"6.3 Cross reference of classification. Military Specification MIL-M-28787/20 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833407. Part number M28787/20-1 supersedes part numbers 2833407 and M28787/20."

Following 6.3, add new 6.4 as follows:

"6.4 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4)."

PAGE 14

FIGURE 5: Delete and substitute new figure 5 as printed on page 5 of this amendment.

PAGE 18

FIGURE 7: Delete and substitute new figure 7 as printed on page 6 of this amendment.

PAGE 21

FIGURE 8: Delete and substitute new figure 8 as printed on page 7 of this amendment.

PAGE 22

* FIGURE 9: Delete and substitute new figure 9 as printed on page 8 of this amendment.

PAGE 53

FIGURE 22: Delete and substitute new figure 22 as printed on page 9 of this amendment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:29Z
Check the source to verify that this is the current version before use.

TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A18	V_{OUT} (low-loaded)	III	S	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
B1-B18	Input breakdown current	IV	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
C1-C18	Input clamp diode voltage	V	Q	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D18	Output voltage, low level	VI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
E1-E18	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F18	Input current, low level	VIII	X	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	mA
G1-G18	Input current, high level	IX	X	0	30	0	40	0	40	0	40	0	40	µA
H1-H18	Output short circuit current	X	X	20	55	20	55	20	55	20	55	20	55	mA
J1	Power supply current, low level	XI	S	1	99	1	99	1	99	1	99	1	99	mA
J2	Power supply current, high level	XI	S	1	36	1	36	1	36	1	36	1	36	mA
K1-K18	Transition time (t_{TLH})	XII	S	4	45	4	50	4	50	4	50	4	50	ns
L1-L18	Transition time (t_{THL})	XII	S	4	20	4	25	4	25	4	25	4	25	ns
M1-M18	Propagation delay time (t_{PHL})	XIII	S	2	19	2	22	2	22	2	22	2	22	ns
N1-N18	Propagation delay time (t_{PLH})	XIII	S	4	28	4	30	4	30	4	30	4	30	ns
P1-P6, P13-P18	Output crosstalk, low level	XIV	Q	0	970	0	970	0	970	0	970	0	970	mV
P7-P12	Output crosstalk, low level	XIV	Q	0	870	0	870	0	870	0	870	0	870	mV
P19-P36	Output crosstalk, low level	XIV	Q	0	420	0	420	0	420	0	420	0	420	mV

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:29Z
Check the source to verify that this is the current version before use.

TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R1-R18	Output crosstalk, high level	XV	Q	0	400	0	400	0	400	0	400	0	400	mV
S	Power supply transient amplitude	XVI	Q	0	600	0	600	0	600	0	600	0	600	mV
T	Power supply transient rise time	XVII	S	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	ms
U1	Isolation													
	Initial	XVIII	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	XVIII	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
U2, U3	Isolation, unused pins													
	Initial	XVIII	Q	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	XVIII	Q	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
V1-V18	Input capacitance	XIX	Q	2	15	2	15	2	15	2	15	2	15	pF
W1-W36	Output impedance, power off, check	XX	Q	3.5	5.55	3.5	5.55	3.5	5.55	3.5	5.55	3.5	5.55	V

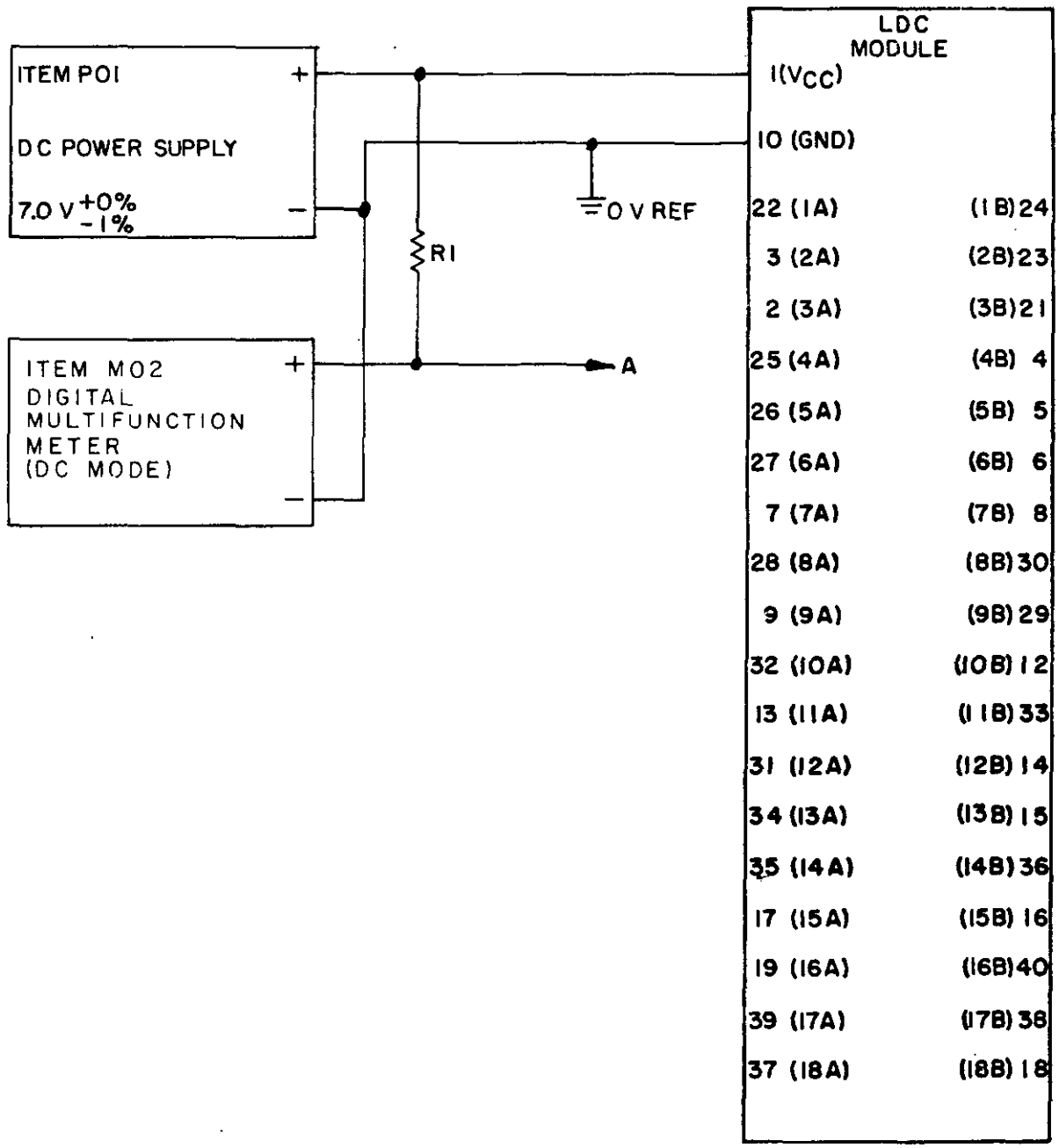
1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

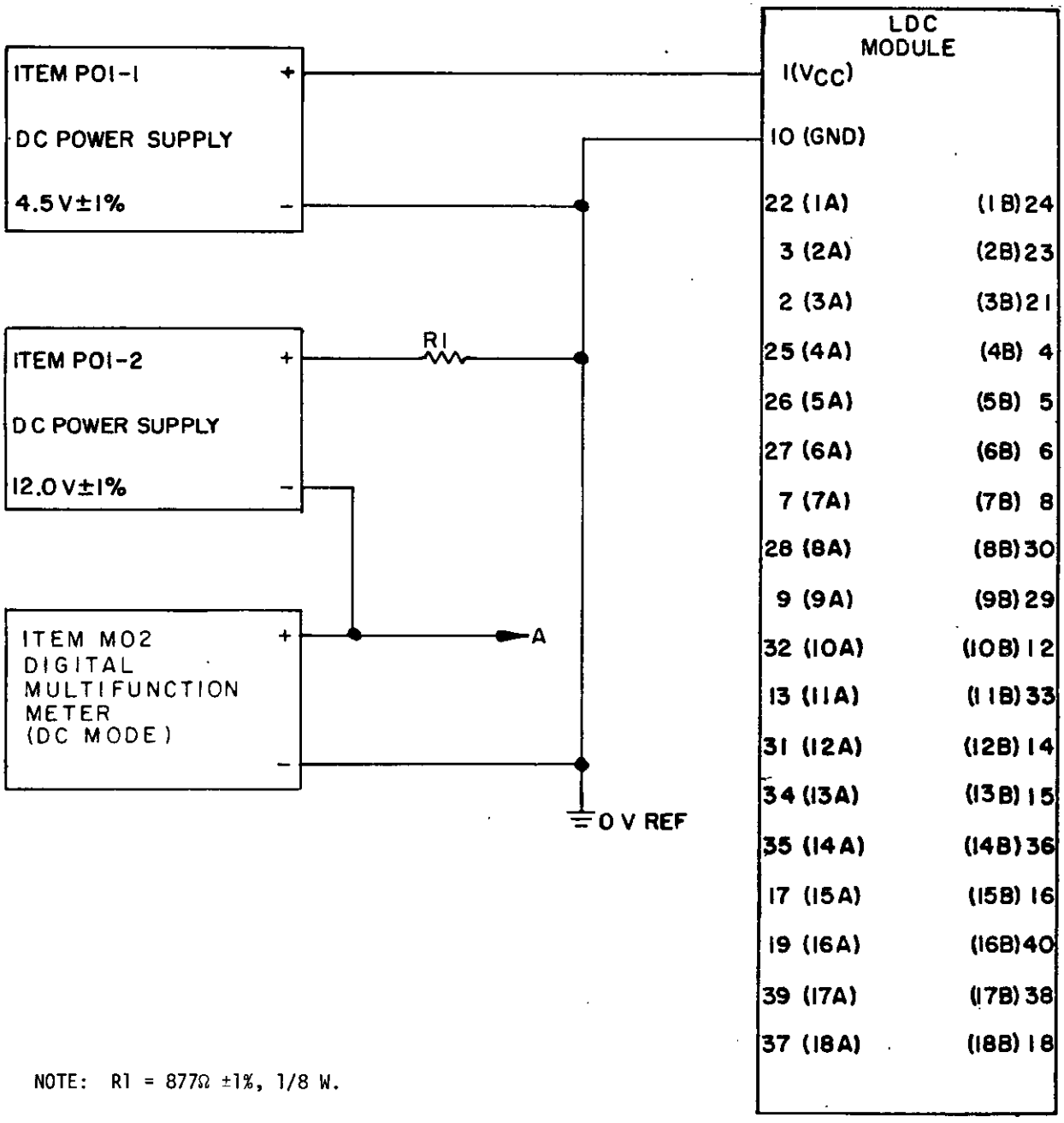
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:29Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 316Ω ±1%, 1/4 W.
2. Measurements shall be recorded after point A has been connected to the appropriate module pin for at least 5 seconds.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

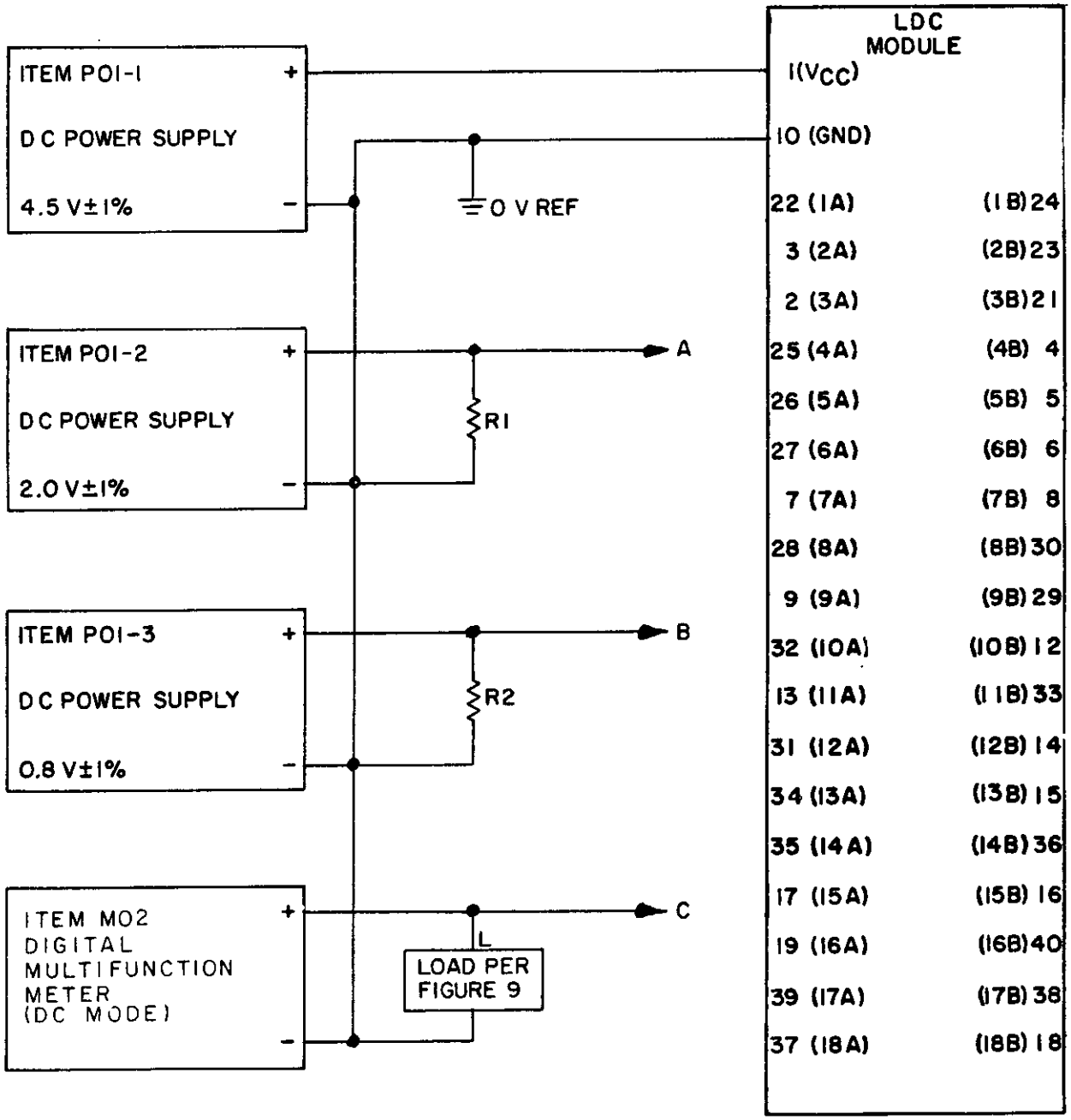
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:29Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 877Ω ±1%, 1/8 W.

FIGURE 7. Input clamp diode voltage test circuit for table V.

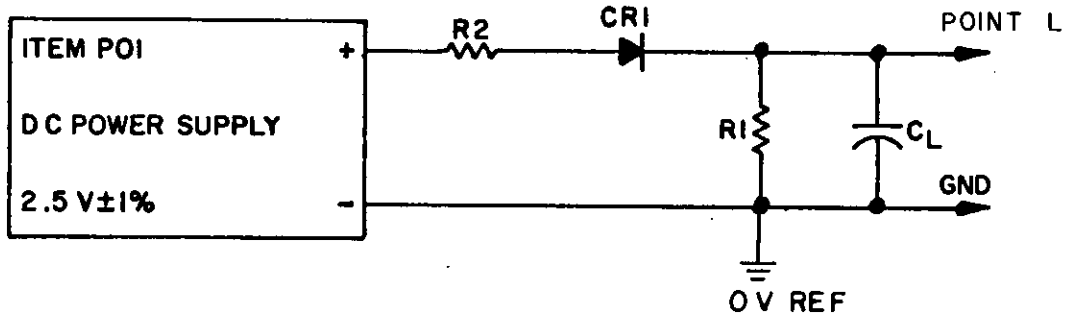
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:29Z
Check the source to verify that this is the current version before use.



NOTE: R1 = R2 = 51Ω ±5%, 1 W.

FIGURE 8. Output voltage (low-level and high-level) test circuit for tables VI and VII.

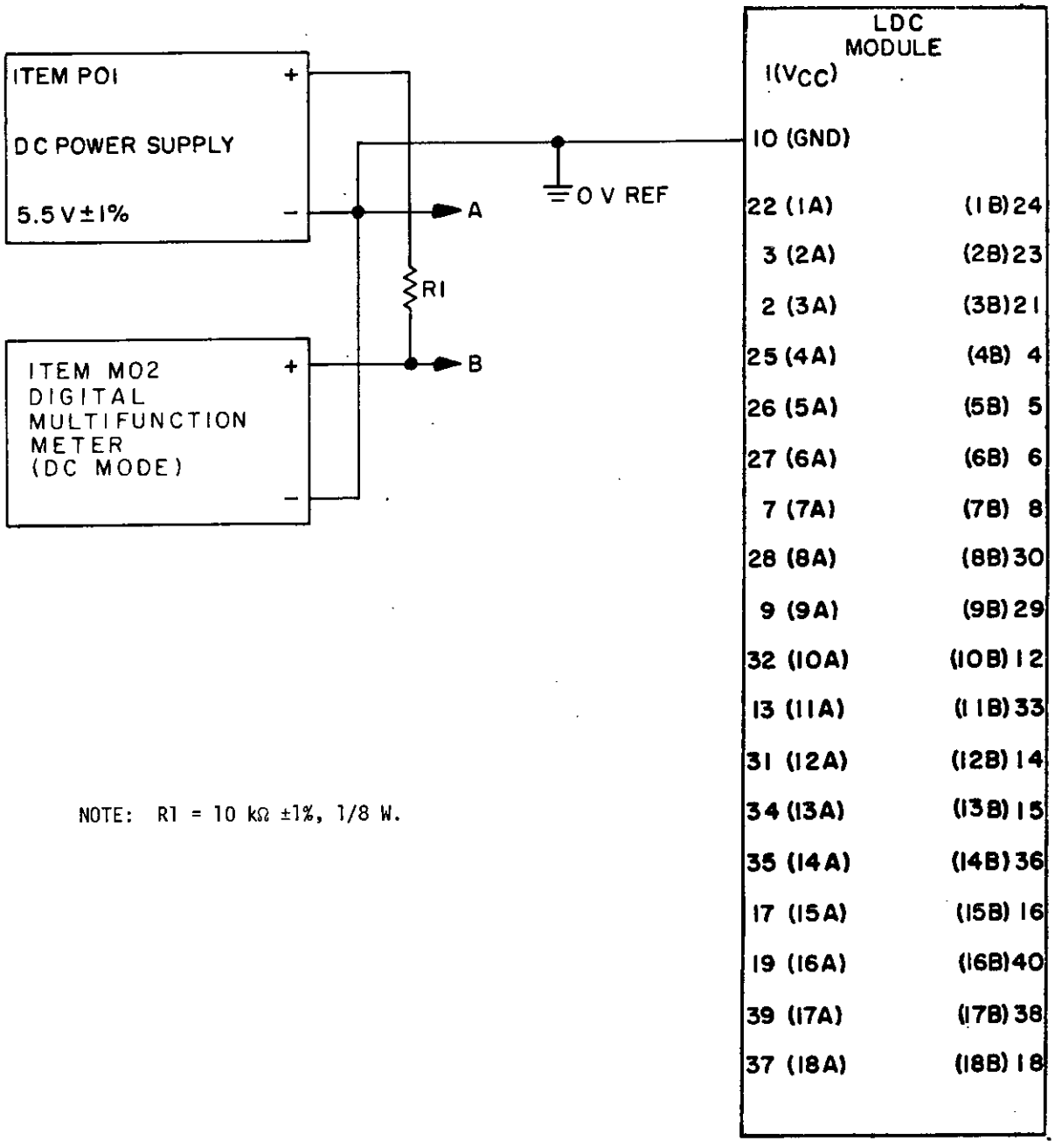
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:29Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 2.43 kΩ ±1%, 1/4 W, metal film.
2. R2 = Select value for a current flow of 16 mA ±5% out of point L with point L held at 0.4 V.
3. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions K, L, M, and N.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions P and R.
5. CR1 = 1N4150, or equivalent.
6. C_L is not required for dc electrical tests D and E.
7. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2 mm).
8. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

*FIGURE 9. Logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:29Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 10 kΩ ±1%, 1/8 W.

FIGURE 22. Output impedance, power off, check test circuit for table XX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:29Z
Check the source to verify that this is the current version before use.

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5963-N683)

*U.S. GOVERNMENT PRINTING OFFICE: 1986—605-036/41848

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:29Z
Check the source to verify that this is the current version before use.